

Multi-Stage Retrieval for BDI-II Symptom Ranking in Social Media: eRisk 2025 Task 1

3-cfu Project Work

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Abstract

Depression is a important contributor to global disability, affecting a large proportion of the population while frequently remaining undiagnosed as a result of the limited awareness of its symptoms. The expanding utilization of social media platforms, including Reddit, offers a valuable source of user-generated text for the identification of indicators of mental health conditions. The present study addresses eRisk 2025 Task 1, which involves ranking social media posts based on their relevance to the 21 symptoms defined in the Beck Depression Inventory-II (BDI-II). We provide a multi-stage retrieval framework to address this issue. To effectively retrieve candidate documents, a BM25-based baseline is first constructed using SQLite FTS5. Second, the all-MiniLM-L6-v2 sentence transformer is used to score semantic similarity in a retrieve-and-rerank method. Third, a hybrid model is presented that combines normalized lexical and semantic scores. Lastly, to improve contextual relevance, a large language model-based reranker called Mistral-7B-Instruct is used with few-shot prompting and 4-bit quantization. The study's findings show that the suggested methods consistently increase performance, with the LLM-based reranker producing the most significant outcomes MAP: 0.0572, P@10: 0.6905, nDCG@10: 0.7097. These results demonstrate the efficacy of combining large language models and semantic models with traditional information retrieval methods for extensive mental health text analysis.

1 Introduction

Depression is a major cause of disability and one of the most common mental health conditions in the world. There is increasing interest in using social media as an additional source of evidence for mental health analysis since many people do not specifically express their psychological condition

in medical settings (Adhikary et al., 2025). Platforms like Reddit contain large amount of naturally occurring, self-reported material which may reveal emotional distress, cognitive patterns, and behavioral symptoms linked to depression. It is difficult to find such signals on a wide scale since significant information is usually indirect, sparse, and embedded in extremely huge collections of noisy user-generated content (Benloucif et al., 2025).

This work addresses eRisk 2025 Task 1 (Parapar et al., 2025), which focuses on ranking social media posts according to their relevance to the 21 symptoms of depression defined in the Beck Depression Inventory-II (BDI-II). The task is significant because it facilitates development of retrieval systems that can extract evidence related to symptoms from large text collections, allowing for downstream analysis in applications linked to mental health. This issue can be solved using a variety of retrieval techniques. Traditional lexical retrieval techniques, like BM25, are effective and have solid empirical support, but they may overlook semantically similar expressions since they rely on precise word overlap. Although sentence embedding-based neural retrieval techniques can capture deeper semantic similarities, they are more computationally costly and may still have difficulty comprehending domain-specific language.

Large language models (LLMs) have demonstrated potential in reranking and contextual relevance estimates in the past few years, but their deployment necessitates careful prompting and resource awareness. We integrate these complementary strengths into a multi-stage retrieval paradigm in this work.

Four systems are used in our approach: (1) a BM25 baseline using SQLite FTS5; (2) a retrieve-and-rerank system using all-MiniLM-L6-v2; (3) a hybrid model linearly combining normalized lexical and semantic scores; and (4) a Mistral-7B-Instruct re-ranker with few-shot prompting and 4-

bit quantization. On a large-scale corpus, this process was created to balance retrieval accuracy, semantic comprehension, and computational viability. The LLM-based reranker achieved MAP of 0.0572, P@10: 0.6905, and nDCG@10: 0.7097, which reflect significant increases over the BM25 baseline across all measures. Experiments conducted on the eRisk 2025 Task 1 dataset demonstrate that ranking quality improves with each subsequent step.

2 Background

Depression is a mental condition characterized by low affect, anhedonia, and a variety of cognitive and physical symptoms. The World Health Organization estimates that it affects over 280 million people globally, making it a major contributor to years spent disabled (Salas-Zárate et al., 2022). Despite being common, depression is often underdiagnosed in clinical settings. This is due in part to patients’ varying symptom presentations and in part to people’s inability to express their distress during medical consultations (Ta et al., 2025).

Social media’s widespread use has created an additional channel for monitoring mental health. Users voluntarily share their experiences in informal, unstructured language in large groups on websites like Reddit. Features from social media writing, such as sentiment, cognitive distortion patterns, and linguistic style, have been shown in earlier computational psychiatry research to be proxies for clinical severity. The eRisk project was created to promote early risk detection via longitudinal social media histories. It is held annually inside the CLEF evaluation framework (Parapar et al., 2025).

The BDI-II is a 21-item self-report tool used in clinical practice to measure the presence and severity of depression symptoms within the previous two weeks (Gebrie, 2018). Each question relates to a certain symptom domain, such as depression, pessimism, loss of enjoyment, suicidal thoughts, exhaustion, and difficulty focusing. Instead of categorizing a user as sad or not, eRisk 2025’s organized, symptom-level granularity gives it an ideal foundation for information retrieval. In order to provide fine-grained evidence, Task 1 asks systems to rank posts according to their relevance to each of the 21 BDI-II symptoms separately.

The retrieve-then-rerank paradigm is commonly used in modern information retrieval for large-scale corpora, driven by the requirement to strike a balance between ranking quality and computing ef-

ficiency. A more costly but more accurate model then reranks the candidates after a first-stage retrieval quickly reduces a corpus of millions of documents to a manageable candidate set.

The predominant first-stage lexical retrieval function is still BM25. By adding document length normalization and a saturation parameter for term frequency, it expands on TF-IDF and becomes robust across collections with different document lengths (Askari et al., 2023). Sentence Transformers are lightweight encoder models that are fine-tuned utilizing contrastive objectives on large paraphrase and NLI datasets. Reimers2019sbert introduced the MiniLM family (Reimers and Gurevych, 2019). Cosine similarity can be used as a relevance indicator that is resistant to surface-level lexical variance since they transfer sentences into a dense vector space so that semantically related texts are geometrically close. By combining lexical and semantic scores-usually after normalization-into a single ranking function, hybrid retrieval takes advantage of the complementing error characteristics of the two signal kinds.

In order to evaluate passage relevance without task-specific fine-tuning, LLMs have lately been used as zero-shot and fewshot rerankers, utilizing their extensive world knowledge and ability to follow instructions. 4-bit quantization, which minimizes generation quality degradation while reducing memory footprint, makes it possible to deploy such models on consumer-grade hardware.

3 System description

Figure 1 illustrates the full multi-stage retrieval pipeline.

3.1 Preprocessing

All 6,300 user TREC files are parsed using a custom streaming parser (`parse_trec_file_stream`), which extracts the <DOCNO>, <TEXT>, <PRE>, and <POST> fields line by line without loading entire files into memory. This design is essential given the scale of 17.5 million posts.

The three text fields are concatenated and passed through a preprocessing function. The resulting cleaned text is then ingested into a SQLite FTS5 virtual table for full-text search, and simultaneously stored in a plain `posts_lookup` table to enable fast content retrieval.

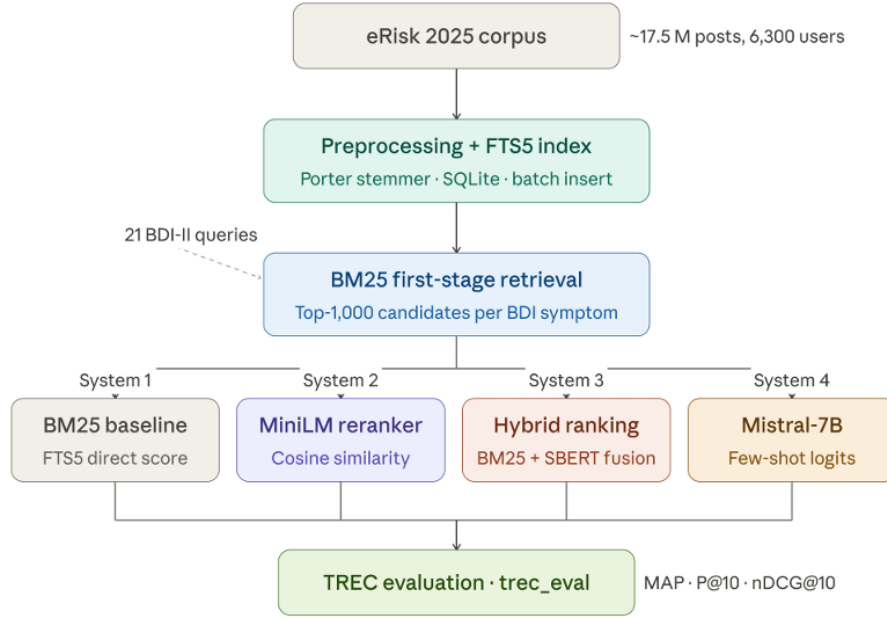


Figure 1: Multi-stage retrieval pipeline

3.2 BM-25 Baseline

A multi-keyword query string covering the vocabulary space of each of the 21 BDI-II symptoms was manually created. For example, the query for *sadness* included: “sad unhappy miserable sorrow depressed down low spirits gloomy.”

At retrieval time, each query is tokenized and redefined as a SQLite FTS5 OR query over individually quoted phrases. By convention, the FTS5 `bm25` function yields negative scores, which are then negated to produce a standard descending ranking. The top 1,000 pages for each query are written in TREC run-file format.

3.3 MiniLM semantic reranker

Each query’s top 1,000 BM25 candidates are sent to `all-MiniLM-L6-v2`, a lightweight six-layer transformer optimized for dense retrieval. Both the candidate text and the query string are encoded into normalized 384-dimensional embeddings, and cosine similarity is used to reorder the candidates.

Encoding is performed in GPU chunks of 128. The final ranking completely replaces the initial BM25 order.

3.4 Hybrid ranking

The final ranking is obtained as a linear combination of lexical and semantic scores. Both score vectors, derived from the same top 1,000 BM25 candidates, are independently min-max normalized

to the range $[0, 1]$ and combined as:

$$\text{hybrid} = 0.2 \times \text{BM25}_{\text{norm}} + 0.8 \times \text{SBERT}_{\text{norm}}$$

The 0.2/0.8 weighting was chosen to de-emphasize exact term overlap (which BM25 already handles in Stage 1) while giving most weight to semantic similarity. Candidates are then ranked in descending order of the hybrid score.

3.5 Mistral-7B LLM reranker

`Mistral-7B-Instruct-v0.3` is configured with 4-bit NF4 quantization (double quantization enabled, float16 computation dtype) to fit within 16 GB of GPU RAM. For each query, the BM25 top 1,000 candidates are ranked by asking the model to determine whether each post is pertinent to the corresponding BDI-II symptom.

The system employs a forward-pass-only scoring method rather than analyzing the results of free-text generation. The tokenized prompt is passed through the model, and the logits at the final real token position are retrieved. A softmax function is applied to the log-probabilities of the two competing first tokens, “relevant” and “not,” producing a continuous relevance probability over only these two positions. Each document thus receives a distinct floating-point score, enabling precise ranking. A user message with the symptom description, four manually written few-shot demonstrations (two relevant, two non-relevant, shuffled for reproducibil-

ity), and the candidate text truncated to 400 characters is sent after a system message announcing the model’s role as an annotator.

4 Data

The dataset is the official eRisk 2025 Task 1 corpus, distributed by the eRisk organizers within the CLEF evaluation framework. It comprises posts from Reddit collected from 6,300 users across diverse subreddits, packaged as 6,300 individual TREC-format files.

After parsing and preprocessing, the full index contains approximately 17.5 million post chunks, with each chunk representing a single `<TEXT>` segment (and associated `<PRE>/<POST>` context) from a user’s posting history. Document IDs follow the format `UserID_PostIndex_ChunkIndex`, enabling reconstruction of per-user posting histories.

A CSV file containing binary relevance labels (*relevant/not-relevant*) for (document, BDI symptom) pairs provided the gold-standard relevance judgments (qrels). Due to significant class imbalance, the qrels cover only a subset of 21 question-specific document pools. Relevant documents constitute a small minority across all 21 inquiries, with a negative-to-positive ratio of roughly 8:1 (the precise ratio varies by query).

For compatibility with `trec_eval`, the qrels were transformed into TREC format.

5 Experimental setup and results

5.1 Setup

Four systems were evaluated in order of increasing complexity. For both the BM25 baseline and the MiniLM re-ranker, all 1,000 BM25 candidates were retained for each query.

For the Hybrid approach, combination weights were heuristically selected to prioritize semantic coherence while preserving a degree of lexical signal, with $\alpha_{\text{BM25}} = 0.2$ and $\alpha_{\text{SBERT}} = 0.8$. No grid search was conducted due to the computational cost associated with running `Mistral-7B-Instruct-v0.3` inference over the full candidate pool.

All sentence embeddings were computed in `float32` precision using `all-MiniLM-L6-v2`. For System 4, `Mistral-7B-Instruct-v0.3` was loaded in `NF4` 4-bit precision via

`BitsAndBytesConfig()`, with a batch size of 4 prompts per GPU pass and a maximum sequence length of 2,048 tokens.

5.2 Evaluation protocol

All four systems generate TREC-formatted run files, each containing the top 1,000 documents per query. Evaluation is performed using `trec_eval`, reporting Mean Average Precision (MAP), Precision at 10 (P@10), and normalized Discounted Cumulative Gain at cut-off 10 (nDCG@10). The `-J` flag is used to treat unjudged documents as non-relevant.

Macro-averaged scores are computed across all 21 BDI-II queries to provide an overall performance comparison.

5.3 Results

Table 1 summarizes the performance of all four systems

Table 1: Retrieval performance across four systems on the eRisk 2025 Task 1 dataset.

System	MAP	P@10	nDCG
BM25 baseline	0.0515	0.5905	0.6240
MiniLM reranker	0.0537	0.6238	0.6384
Hybrid Model	0.0535	0.6238	0.6394
Mistral-7B	0.0572	0.6905	0.7097

As the pipeline progresses from purely lexical retrieval to semantic and hybrid models, and finally to LLM-based reranking, the results show consistent improvement across all three evaluation metrics. Compared to the BM25 baseline, the **Mistral-7B** reranker achieves the largest absolute gains, increasing MAP from 0.0515 to 0.0572 (an 11% relative improvement), nDCG@10 from 0.6240 to 0.7097 (a 13.7% relative improvement), and P@10 from 0.5905 to 0.6905 (a 16.9% relative improvement).

It is notable that all systems have low absolute MAP levels. This illustrates the task’s inherent sparsity: MAP is structurally suppressed because only a small portion of the 17.5 million postings were deemed relevant per symptom, and many pertinent papers are probably missing from the retrieved top-1,000 pool (recall ceiling). A more detailed picture of practical retrieval quality is given by P@10 and nDCG@10, which concentrate on the quality of the top-ranked results.

6 Discussion

MAP and nDCG@10 consistently improve from BM25 to semantic and hybrid systems, demonstrating that semantic reranking has benefits beyond precise keyword matching. MAP is raised by the MiniLM reranker from 0.0515 to 0.0537 (+4.3%) and nDCG@10 from 0.6240 to 0.6384 (+2.3%). While MAP is almost unaltered in comparison to MiniLM alone, the hybrid system obtains a slightly better nDCG@10 of 0.6394, a modest further gain. According to this pattern, the majority of the semantic signal is already captured by MiniLM, and the hybrid's BM25 component mainly reinforces exact keyword matches that the semantic model may downweight.

Performance varied markedly across the 21 BDI queries. Characteristic failure modes are revealed by examining the top five results for each system for certain BDI queries. For BM25 (Query 21: Loss of Interest in Sex), postings that commonly utilize the phrase "intimacy" in regard to relationship advice rather than in relation to personal symptom experiences tend to rank highest. This implies that symptoms with a wide everyday vocabulary are not sufficiently discriminated by the FTS5 query expansion (keyword bag).

The MiniLM reranker addresses this in part by rewarding posts with a semantic vector that is closer to the BDI symptom description, but it occasionally retrieves very short, semantically vacuous posts (e.g., "it is only human to crave intimacy") simply because their embedding is close to the query. There is no clinically significant information in these posts.

In contrast, the Mistral-7B reranker more closely aligns with the BDI-II evaluation purpose by highlighting posts in which the user openly explains their personal experience (e.g., "before, my desire was basically gone"). The model's sensitivity to the aboutness distinction may not be adequately calibrated by the few-shot demonstrations alone, as Mistral sometimes gives high relevance scores to postings that are framed in the first person but only tangentially reference a symptom.

7 Conclusion

In conclusion, for the eRisk 2025 Task 1 model, we implemented a four-stage retrieval pipeline to evaluate social media posts according to their relation to BDI-II depressive symptoms. Each step BM25 lexical retrieval, MiniLM semantic rerank-

ing, hybrid fusion, and Mistral-7B few-shot reranking improves ranking quality in a quantifiable way by building on the previous stage. The best overall results (MAP = 0.0572, P@10 = 0.6905, nDCG@10 = 0.7097) are obtained by the LLM-based reranker, demonstrating the usefulness of contextual, instruction-following models for complex relevance tasks in the field of mental health.

The pipeline's primary weakness is its reliance on BM25 for first-stage candidate selection: relevant postings that employ non-overlapping vocabulary are permanently omitted from downstream reranking. Future research should examine domain-adaptive fine-tuning of the sentence transformer on depression-related text to better calibrate symptom-level similarity and investigate dense first-stage retrieval (FAISS indexed SBERT embeddings) to increase memory. The author-aboutness discrimination, which is essential for BDI-II relevance, may be further enhanced by adding chain-of-thought prompting for Mistral and broadening the few-shot example pool.

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